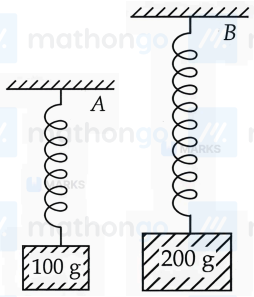
Q1. JEE Main 2026 (21 January Shift 2)

The kinetic energy of a simple harmonic oscillator is oscillating with angular frequency of $176 \mathrm{rad/s}$. The frequency of this simple harmonic oscillator is ____ Hz. [take $\pi = \frac{22}{7}$]

(1) 14 (2) 28 (3) 176 (4) 88

Q2. JEE Main 2026 (23 January Shift 1)

Two blocks with masses 100 g and 200 g are attached to the ends of springs A and B as shown in figure. The energy stored in A is E. The energy stored in B, when spring constants k_A, k_B of A and B, respectively satisfy the relation $4k_A = 3k_{B'}$, is :



(1) $\frac{4}{3}E$ (2) 4E (3) 3E (4) $2E$

Q3. JEE Main 2026 (23 January Shift 1)

A simple pendulum of string length 30 cm performs 20 oscillations in 10 s. The length of the string required for the pendulum to perform 40 oscillations in the same time duration is ____ cm. [Assume that the mass of the pendulum remains same.]

(1) 7.5 (2) 0.75 (3) 120 (4) 15

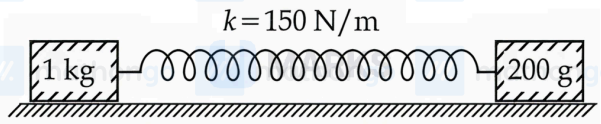
Q4. JEE Main 2026 (28 January Shift 2)

The time period of a simple harmonic oscillator is $T = 2\pi\sqrt{\frac{k}{m}}$. Measured value of mass (m) of the object is 10 g with an accuracy of 10 mg and time for 50 oscillations of the spring is found to be 60 s using a watch of 2 s resolution. Percentage error in determination of spring constant (k) is ____ \%.

(1) 7.60 (2) 3.35 (3) 3.43 (4) 6.76

Q5. JEE Main 2026 (28 January Shift 2)

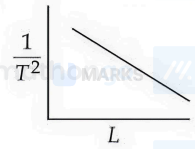
As shown in the figure, a spring is kept in a stretched position with some extension by holding the masses 1 kg and 0.2 kg with a separation more than spring natural length and are released. Assuming the horizontal surface to be frictionless, the angular frequency (in SI unit) of the system is : $k = 150 \mathrm{~N/m}$

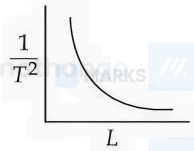


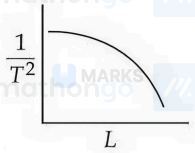
(1) 5 (2) 27 (3) 20 (4) 30

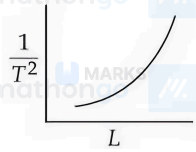
Q6. JEE Main 2026 (22 January Shift 2)

Using a simple pendulum experiment g is determind by measuring its time period T. Which of the following plots represent the correct relation between the pendulum length L and time period T ?

(1) 

(2) 

(3) 

(4) 

Q7. JEE Main 2026 (28 January Shift 1)

The displacement of a particle, executing simple harmonic motion with time period T, is expressed as $x(t) = A\sin\omega t$, where A is the amplitude. The maximum value of potential energy of this oscillator is found at $t = T/2\beta$. The value of β is ____.

Q8. JEE Main 2026 (24 January Shift 1)

A cylindrical block of mass M and area of cross section A is floating in a liquid of density ρ and with its axis vertical. When depressed a little and released the block starts oscillating. The period of oscillation is ____.

(1) $\pi\sqrt{\frac{2M}{\rho Ag}}$ (2) $\pi\sqrt{\frac{\rho A}{Mg}}$ (3) $2\pi\sqrt{\frac{\rho A}{Mg}}$ (4) $2\pi\sqrt{\frac{M}{\rho Ag}}$

ANSWERS AND SOLUTIONS

1. (1) **2.** (3) **3.** (1) **4.** (4) **5.** (4) **6.** (2) **7.** 2 **8.** (4)

1. **(1)** In a simple harmonic oscillator, if the displacement is $x = A\sin(\omega t + \phi)$, the velocity is $v = A\omega\cos(\omega t + \phi)$.
The kinetic energy is given by $K = \frac{1}{2}mv^2 = \frac{1}{2}mA^2\omega^2\cos^2(\omega t + \phi)$.
Using the identity $\cos^2\theta = \frac{1+\cos 2\theta}{2}$, we get $K = \frac{1}{4}mA^2\omega^2[1 + \cos(2\omega t + 2\phi)]$.
The angular frequency of the kinetic energy oscillation is $\omega_{KE} = 2\omega$, where ω is the angular frequency of the SHM.
Given $\omega_{KE} = 176$ rad/s, we have $2\omega = 176$, which gives $\omega = 88$ rad/s.
The frequency of the simple harmonic oscillator is $f = \frac{\omega}{2\pi}$.
Substituting the values: $f = \frac{88}{2\times\frac{22}{7}} = \frac{88\times 7}{44} = 2 \times 7 = 14$ Hz.

2. **(3)** For blocks hanging in equilibrium, spring extensions are:
$x_A = \frac{m_A g}{k_A} = \frac{100g}{k_A}$ and $x_B = \frac{m_B g}{k_B} = \frac{200g}{k_B}$.
Energy stored:
$E = \frac{1}{2}k_A x_A^2 = \frac{5000g^2}{k_A}$ and $U_B = \frac{1}{2}k_B x_B^2 = \frac{20000g^2}{k_B}$.
Taking ratio: $\frac{U_B}{E} = 4 \cdot \frac{k_A}{k_B}$.
Given $4k_A = 3k_B$, we have $\frac{k_A}{k_B} = \frac{3}{4}$.
Therefore: $U_B = 4E \cdot \frac{3}{4} = 3E$.

3. **(1)** Period of simple pendulum: $T = \frac{\text{time}}{\text{number of oscillations}}$
Initial period: $T_1 = \frac{10}{20} = 0.5$ s
Required period: $T_2 = \frac{10}{40} = 0.25$ s
Using $T = 2\pi\sqrt{\frac{L}{g}}$, the ratio is: $\frac{T_2}{T_1} = \sqrt{\frac{L_2}{L_1}}$
$\frac{0.25}{0.5} = \sqrt{\frac{L_2}{30}} \Rightarrow 0.5 = \sqrt{\frac{L_2}{30}}$
$0.25 = \frac{L_2}{30} \Rightarrow L_2 = 7.5$ cm

4. **(4)** The given formula for time period is
$T = 2\pi\sqrt{\frac{k}{m}}$.
Squaring both sides, we get $T^2 = 4\pi^2\frac{k}{m}$, which gives $k = \frac{4\pi^2 m}{T^2}$.
The relative error in k is given by $\frac{\Delta k}{k} = \frac{\Delta m}{m} + 2\frac{\Delta T}{T}$.
Given: $m = 10$ g, $\Delta m = 10$ mg $= 0.01$ g.
Total time for $n = 50$ oscillations is $t = 60$ s with resolution $\Delta t = 2$ s.
Since $T = t/n$, the relative error in T is the same as in t: $\frac{\Delta T}{T} = \frac{\Delta t}{t}$.
Substituting the values: $\frac{\Delta k}{k} = \frac{0.01}{10} + 2\left(\frac{2}{60}\right)$.
$\frac{\Delta k}{k} = 0.001 + \frac{2}{30} = 0.001 + 0.0666...$

Percentage error $= \left(0.001 + \frac{1}{15}\right) \times 100 = 0.1 + \frac{100}{15} = 0.1 + 6.666... = 6.766...\%$.
Rounding to two decimal places, we get 6.76%.

5. **(4)** Two masses (1 kg and 0.2 kg) connected by a spring (k = 150 N/m) on a frictionless surface undergo oscillation about their center of mass.
This two-body problem reduces to SHM with reduced mass $\mu = \frac{m_1 m_2}{m_1+m_2} = \frac{1\times 0.2}{1.2} = \frac{1}{6}$ kg.
The angular frequency is $\omega = \sqrt{\frac{k}{\mu}} = \sqrt{\frac{150}{1/6}} = \sqrt{900} = 30$ rad/s.

6. **(2)** The time period T of a simple pendulum of length L is given by the formula:
$T = 2\pi\sqrt{\frac{L}{g}}$
Squaring both sides, we get:
$T^2 = 4\pi^2 \frac{L}{g}$
Rearranging the equation to find the relationship between $\frac{1}{T^2}$ and L:
$\frac{1}{T^2} = \frac{g}{4\pi^2 L}$
Let $y = \frac{1}{T^2}$ and $x = L$. The equation becomes:
$y = \frac{k}{x}$ where $k = \frac{g}{4\pi^2}$ is a constant.
This is the equation of a rectangular hyperbola in the first quadrant.
As L increases, $\frac{1}{T^2}$ decreases non-linearly, approaching the axes asymptotically.
Comparing this with the given plots, option (2) correctly represents a hyperbolic relationship between $\frac{1}{T^2}$ and L.

7. **(2)** For SHM with $x(t) = A\sin(\omega t)$, the potential energy is
$U = \frac{1}{2}m\omega^2 x^2 = \frac{1}{2}m\omega^2 A^2 \sin^2(\omega t)$.
Maximum potential energy occurs when $\sin^2(\omega t) = 1$, i.e., when $\omega t = \frac{\pi}{2}, \frac{3\pi}{2}, \ldots$
The first maximum occurs at $\omega t = \frac{\pi}{2}$, giving $t = \frac{\pi}{2\omega} = \frac{\pi}{2} \cdot \frac{T}{2\pi} = \frac{T}{4}$.
Since this is given as $t = \frac{T}{2\beta}$.
We have $\frac{T}{4} = \frac{T}{2\beta}$, so $\beta = 2$.

8. **(4)** When the block is depressed by distance x, the additional buoyant force is $F = \rho A g x$ acting upward as restoring force.
Equation of motion: $M\frac{d^2x}{dt^2} = -\rho A g x$.
This gives $\omega^2 = \frac{\rho A g}{M}$.
Period of oscillation: $T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{M}{\rho A g}}$